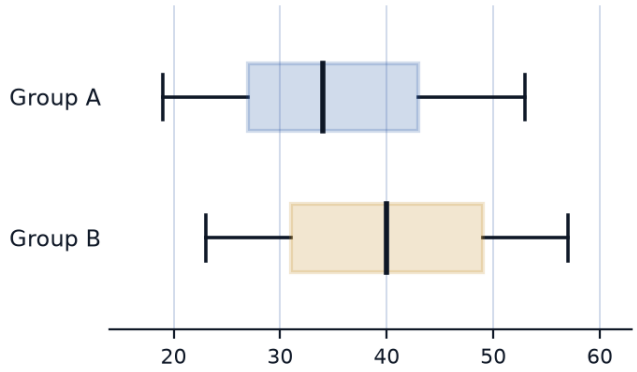
Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 [Foundational] • Target: Measures of center

Find the mean of $12, 15, 18, 9, 21$.

2 [Foundational] • Target: Measures of center

Find the median of $4, 9, 2, 11, 7, 15$.



3 [Foundational] • Target: Measures of center

Find the mode of $3, 7, 5, 7, 2, 7, 9$.

4 [Foundational] • Target: Measures of spread

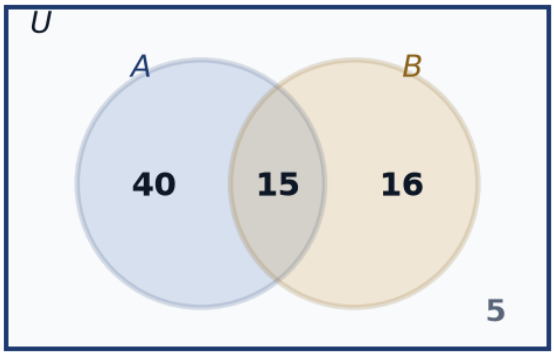
Find the range of the data set $22, 18, 35, 9, 27$.

5 [Foundational] • Target: Measures of spread

Find the interquartile range (IQR) of $4, 7, 9, 12, 15, 18, 21$.

6 [Foundational] • Target: Measures of spread

A data set has a range of 40 and a minimum value of 15. Find the maximum value.



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7 [Foundational] • Target: Basic probability

A jar has 5 red, 3 blue, and 2 green marbles. Find $P(\text{green})$.

Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 [Intermediate] • Target: Basic probability

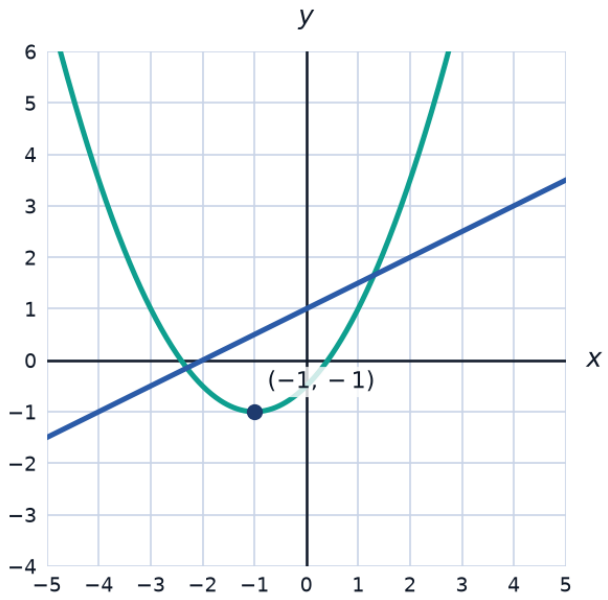
A fair six-sided die is rolled once. Find $P(\text{even number})$.

9 [Intermediate] • Target: Basic probability

A spinner has 8 equal sections numbered 1 through 8. Find $P(\text{number greater than } 5)$.

10 [Intermediate] • Target: Combined probability: AND / OR

A card is drawn from a standard 52-card deck. Find $P(\text{king or queen})$.



11 [Intermediate] • Target: Combined probability: AND / OR

Two independent events have $P(A) = 0.4$ and $P(B) = 0.5$. Find $P(A \text{ and } B)$.

12 [Intermediate] • Target: Combined probability: AND / OR

A bag has 6 yellow and 4 purple marbles. A marble is drawn, replaced, then a second is drawn. Find $P(\text{both yellow})$.

13 [Intermediate] • Target: Probability without replacement

A bag has 5 red and 7 blue marbles. Two marbles are drawn without replacement. Find $P(\text{both red})$.

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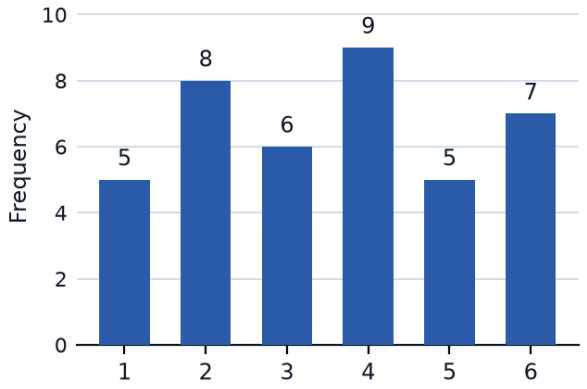
14 [Intermediate] • Target: Probability without replacement

A committee of 2 is chosen at random, one at a time without replacement, from a group of 4 men and 6 women. Find $P(\text{both chosen are women})$.

Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

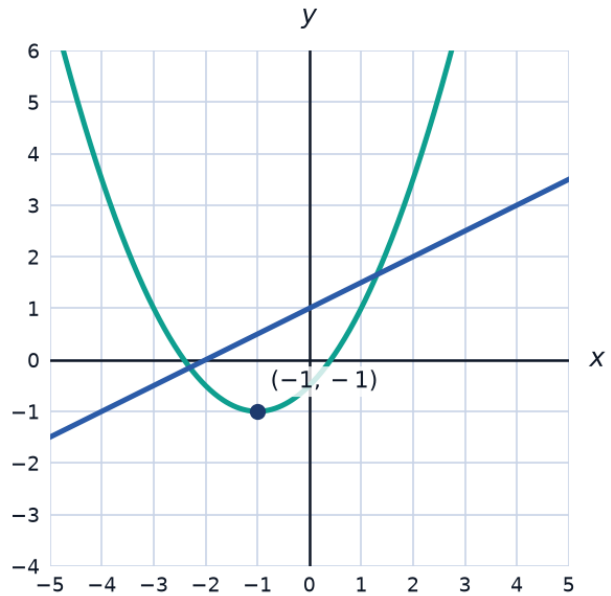
15 [Challenging] • Target: Data displays • Cross-Lesson: 6.01 Geometry

The bar graph shows the results of rolling a fair die 40 times. Based on the experimental data, find the experimental probability of rolling a 4, and compare it to the theoretical probability.



16 [Challenging] • Target: Expected value • Cross-Lesson: 7.01 Trigonometry & Additional Topics

A game costs 2 dollars to play. A player wins 10 dollars with probability 0.1, and wins nothing otherwise. Find the expected net profit per play.



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17 [Challenging] • Target: Expected value • Cross-Lesson: 8.01 Exponentials & Logarithms

A spinner has 4 equal sections labeled 5, 10, 15, and 20 points. Find the expected value of one spin.

18 [Challenging] • Target: Effects of transformations on statistics • Cross-Lesson: 1.01 Heart of Algebra

A data set has mean 20. If every value in the set is increased by 5, find the new mean.

19 [Challenging] • Target: Effects of transformations on statistics • Cross-Lesson: 2.01 Quadratics & Polynomials

A data set has standard deviation 4. If every value is multiplied by 3, find the new standard deviation.

20 [Challenging] • Target: A well-designed word problem: a survey of students • Cross-Lesson: 3.01 Functions & Graphs

This question has three parts. A researcher surveys 200 students about whether they play a sport and whether they play a musical instrument. Of the 200 students, 120 play a sport, 80 play an instrument, and 30 do both. (a) Find how many students do neither activity. (b) Find the probability that a randomly chosen student plays a sport or an instrument (or both). (c) Given that a randomly selected student plays a sport, find the probability that they also play an instrument.

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

21 [Foundational] • Target: Measures of center

Find the mean of $6, 50, 46, 36, 7$.

a. 29

b. 31

c. 145

d. 27

22 [Foundational] • Target: Measures of spread

Find the mean of $21, 18, 14, 35, 2$.

a. 90

b. 16

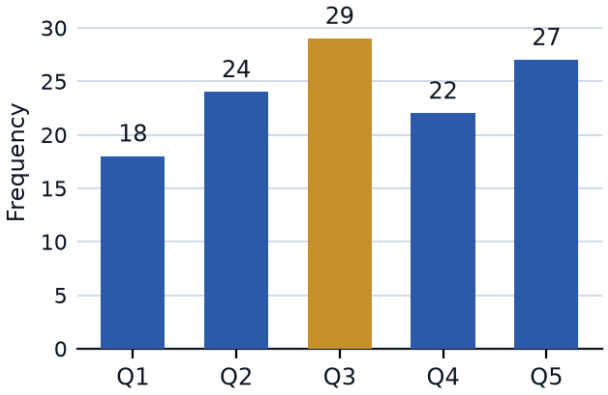
c. 18

d. 20

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23 [Foundational] • Target: Basic probability

Find the mean of 23, 12, 44, 5, 26.



a. 24

b. 20

c. 110

d. 22

24 [Foundational] • Target: Combined probability: AND / OR

Find the mean of 4, 31, 3, 26, 1.

a. 13

b. 11

c. 65

d. 15

25 [Foundational] • Target: Probability without replacement

Find the mean of 9, 21, 19, 19, 22.

a. 18

b. 16

c. 20

d. 90

26 [Foundational] • Target: Data displays

Find the mean of 50, 23, 27, 33, 12.

a. 145

b. 31

c. 29

d. 27

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27 [Foundational] • Target: Expected value

Find the mean of $14, 29, 33, 44, 30$.

a. 28

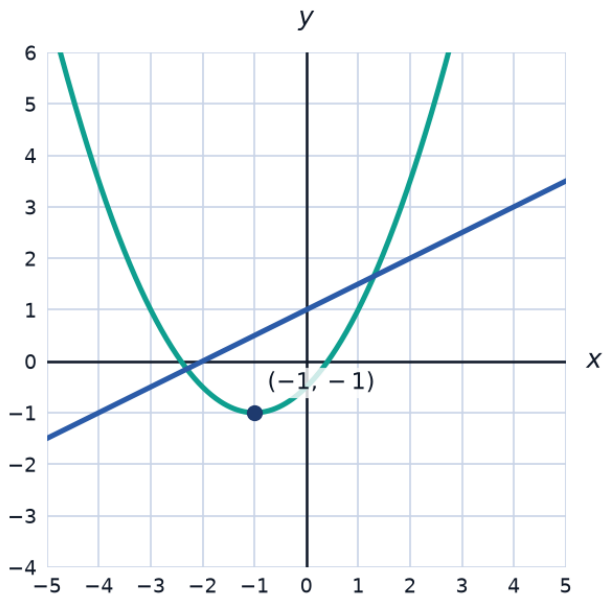
b. 32

c. 30

d. 150

28 [Foundational] • Target: Effects of transformations on statistics

A jar has 11 yellow, 6 red, and 6 green marbles. Find $P(\text{red})$.



a. $\frac{6}{23}$

b. $\frac{8}{23}$

c. $\frac{11}{23}$

d. $\frac{9}{23}$

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

29 [Intermediate] • Target: A well-designed word problem: a survey of students

A jar has 4 red, 8 purple, and 4 green marbles. Find $P(\text{red})$.

a. $\frac{7}{16}$

b. $\frac{3}{8}$

c. $\frac{1}{4}$

d. $\frac{1}{2}$

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30 [Intermediate] • Target: Summary Statistics & Outliers

A jar has 9 red, 2 green, and 10 blue marbles. Find $P(\text{red})$.

a. $\frac{2}{21}$

b. $\frac{10}{21}$

c. $\frac{11}{21}$

d. $\frac{3}{7}$

31 [Intermediate] • Target: Normal Distributions & Standard Deviation

A jar has 2 red, 5 purple, and 1 orange marbles. Find $P(\text{red})$.

a. $\frac{1}{8}$

b. $\frac{1}{2}$

c. $\frac{5}{8}$

d. $\frac{1}{4}$

32 [Intermediate] • Target: Compound & Conditional Probability

A jar has 10 red, 8 purple, and 9 blue marbles. Find $P(\text{red})$.

a. $\frac{1}{3}$

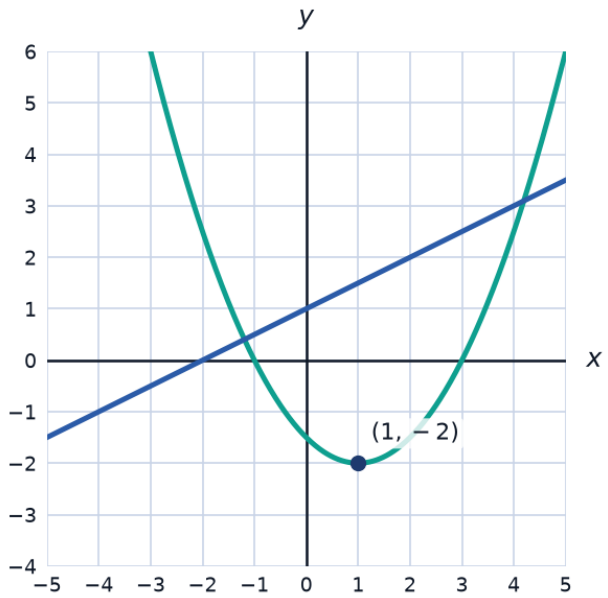
b. $\frac{8}{27}$

c. $\frac{10}{27}$

d. $\frac{4}{9}$

33 [Intermediate] • Target: Permutations & Combinations

A jar has 12 green, 12 orange, and 2 blue marbles. Find $P(\text{blue})$.



a. $\frac{5}{26}$

b. $\frac{1}{13}$

c. $\frac{2}{13}$

d. $\frac{6}{13}$

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34 [Intermediate] • Target: Measures of center

A jar has 9 blue, 10 orange, and 3 red marbles. Find $P(\text{blue})$.

a. $\frac{5}{11}$

b. $\frac{9}{22}$

c. $\frac{3}{22}$

d. $\frac{1}{2}$

35 [Intermediate] • Target: Measures of spread

Two independent events have $P(A) = 0.1$ and $P(B) = 0.7$. Find $P(A \text{ and } B)$.

a. 0.27

b. 0.17

c. 0.12

d. 0.07

36 [Intermediate] • Target: Basic probability

Two independent events have $P(A) = 0.8$ and $P(B) = 0.5$. Find $P(A \text{ and } B)$.

a. 0.40

b. 0.30

c. 0.50

d. 0.60

37 [Intermediate] • Target: Combined probability: AND / OR

A bag has 7 red and 9 blue marbles. Two marbles are drawn without replacement. Find $P(\text{both red})$.

a. $\frac{7}{16}$

b. $\frac{21}{40}$

c. $\frac{3}{10}$

d. $\frac{7}{40}$

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

38 [Challenging] • Target: Probability without replacement • Cross-Lesson: 1.01 Heart of Algebra

A bag has 7 red and 4 blue marbles. Two marbles are drawn without replacement. Find $P(\text{both red})$.

a. $\frac{21}{55}$

b. $\frac{28}{55}$

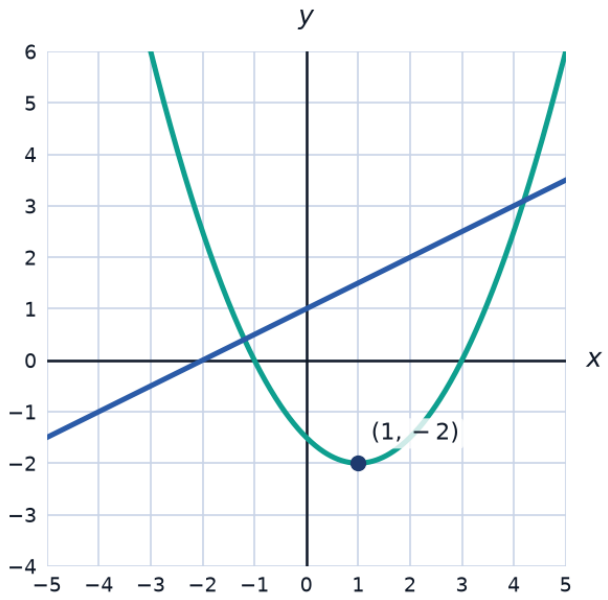
c. $\frac{7}{11}$

d. $\frac{6}{55}$

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39 [Challenging] • Target: Data displays • Cross-Lesson: 2.01 Quadratics & Polynomials

A spinner has 4 equal sections labeled 10, 20, 25, and 40 points. Find the expected value of one spin.



a. 26.75

b. 24.75

c. 23.75

d. 22.75

40 [Challenging] • Target: Expected value • Cross-Lesson: 3.01 Functions & Graphs

A spinner has 4 equal sections labeled 15, 20, 30, and 40 points. Find the expected value of one spin.

a. 27.25

b. 26.25

c. 29.25

d. 25.25

41 [Challenging] • Target: Summary Statistics & Outliers • Cross-Lesson: 4.01 Ratios, Rates & Data

The mean of five numbers is 24. When a sixth number is added, the mean becomes 26. What is the sixth number?

a. 32

b. 34

c. 36

d. 38

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42 [Challenging] • Target: Summary Statistics & Outliers • Cross-Lesson: 6.01 Geometry

An outlier is defined as any data point greater than $Q_3 + 1.5 \times \text{IQR}$. If $Q_1 = 20$ and $Q_3 = 35$, which value is considered an outlier?

a. 50

b. 55

c. 58

d. 60

43 [Challenging] • Target: Normal Distributions & Standard Deviation • Cross-Lesson: 7.01 Trigonometry & Additional Topics

Which change will increase the standard deviation of a dataset?

a. Adding 10 to all data points

b. Multiplying all data points by 2

c. Subtracting the mean from each data point

d. Dividing all data points by 3

44 [Challenging] • Target: Compound & Conditional Probability • Cross-Lesson: 8.01 Exponentials & Logarithms

If $P(A) = 0.5$, $P(B) = 0.6$, and $P(A \cup B) = 0.8$, what is $P(A \cap B)$?

a. 0.1

b. 0.2

c. 0.3

d. 0.4

45 [Challenging] • Target: Permutations & Combinations • Cross-Lesson: 1.01 Heart of Algebra

How many distinct 3-person executive committees (President, Vice-President, Treasurer) can be elected from a club of 10 members?

a. 120

b. 240

c. 720

d. 1000

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